
DATA LOSS PREVENTION (DLP)

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FOR SLIDES, SUPPLEMENTARY LABS,
QUIZZES AND ARTICLES,
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SCENARIO

Ilona is planning to deploy a data loss prevention (DLP) system in her cloud environment. Which of the following challenges is most likely to impact the ability of her DLP system to determine whether sensitive data is being transmitted outside of her organisation?

- A. Lack of data labelling
- B. Use of encryption for data in transit
- C. Improper data labelling
- D. Use of encryption for data at rest

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INTRODUCTION TO DATA CLASSIFICATION

Key topic: The foundation of asset security

Sub topics:

- Importance of identifying and classifying information
- Sensitive data: definitions and examples
- Security policies and asset labelling



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ATTRIBUTES OF SENSITIVE DATA

Definition: Any information
that is not public or
unclassified

Examples:

- Confidential
- Proprietary
- Regulatory-protected data

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DATA CLASSIFICATION AND CATEGORISATION

Key terms:

- **Classification:** Identifying attributes of data
- **Categorisation:** Determining system/data protection requirements

Examples of control expectations:

- *Public data:* No encryption in transit required
- *Internal Use Only:* Encryption in transit and at rest

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FACTORS INFLUENCING CLASSIFICATION LEVELS

Drivers of classification schemes:

- **Data type:** Healthcare, financial, PII
- **Legal constraints:** GDPR compliance, geolocation
- **Ownership:** Third-party requirements
- **Value/Criticality:** Impact on operations

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DATA LABELLING

Importance of labelling:

- Communicating classification levels
- Supporting DLP tools in data identification



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INTRODUCTION TO DATA LOSS PREVENTION

Definition: Security solution to prevent data leakage

Scope:

- Monitors on-premises systems, cloud, and endpoints
- Ensures compliance with regulations

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HOW DLP WORKS

Key components:

- **People:** Training and awareness
- **Processes:** Policies and procedures
- **Technology:** Tools for detection and prevention

Methods :

- Pattern matching
- Policy-based detection

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DLP AND ENCRYPTION/UNENCRYPTED DATA

Role of unencrypted data:

- Enables scanning of keywords and patterns
- Examples: Social security numbers, classification labels

Limitations:

- Encrypted data cannot be scanned directly

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SUMMARY AND BEST PRACTICES

Key takeaways:

- Classify and label data effectively
- Balance encryption and DLP needs
- Regularly update DLP policies and tools



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